

6. Project Testing Phase

1. INTRODUCTION

The Project Testing Phase is the sixth stage in the development of **LegalEase AI**. After completing the development and integration of the frontend, backend, and AI/LLM components, the application needs to be tested to ensure that it works as expected.

In this phase, different user inputs and general legal queries are provided to the application. The generated AI responses, backend APIs, and overall response flow are checked. Any errors, unexpected outputs, or connectivity problems identified during testing are corrected.

The main purpose of this phase is to verify the functionality, usability, API connectivity, and overall performance of the LegalEase AI application.

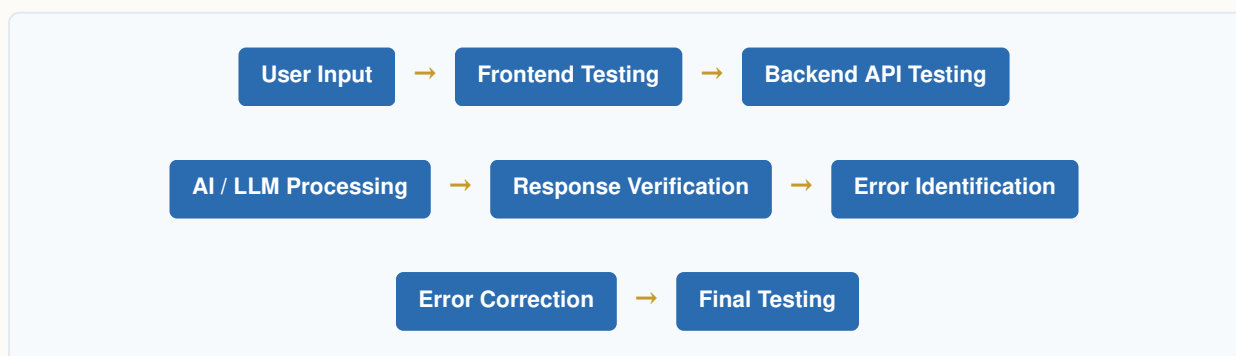
2. OBJECTIVES

The main objectives of the Project Testing Phase are:

- Test the complete LegalEase AI application.
- Check different types of user inputs.
- Verify AI-generated responses.
- Test backend API functionality.
- Check AI/LLM API connectivity.
- Identify errors and unexpected outputs.
- Correct technical issues found during testing.
- Check the overall usability of the application.
- Verify the complete response flow.

3. TESTING WORKFLOW

The testing process follows a systematic sequential flow from client input to final verification:



4. DETAILED TESTING

4.1 User Input Testing

Different types of general legal questions are entered into the application to check whether the system accepts and processes the input correctly. For example, when entering *"What is a rental agreement?"*, the system must accept the query and forward it to the backend.

4.2 Backend API Testing

The backend API is tested to verify whether it correctly receives requests from the frontend. For instance, testing the /ask route using payload:

```
{  
  "question": "What is a rental agreement?"  
}
```

The backend processes the payload and returns an appropriate structured response.

4.3 AI Response Testing

The AI/LLM response is checked to verify whether the system generates an understandable response for the user's question. The response should be relevant, easy to understand, properly displayed, and suitable for general legal informational assistance.

4.4 Error Testing

The application is tested with invalid or incomplete inputs, such as {"question": ""}. The system must identify missing parameters and return an informative error message rather than failing unexpectedly.

4.5 API Connectivity Testing

The connection between the backend and AI/LLM service is verified. This includes checking the API connection, API request format, API response structure, environment configuration, and error handling mechanics.

5. TEST CASES

Test Case	Input	Expected Result
Valid Question	General legal question	AI response generated
Empty Input	Blank question	Error message displayed
API Test	Valid API request	Successful response
AI Test	Legal query	Relevant AI response

Test Case	Input	Expected Result
Multiple Queries	Different legal questions	System processes each query
UI Test	User interaction	Interface works correctly

6. ERROR IDENTIFICATION AND CORRECTION

During testing, potential bottlenecks—such as invalid user input, backend API errors, AI/LLM connectivity disruptions, configuration mismatches, unexpected AI responses, or frontend rendering bugs—are identified. Upon applying code and configuration fixes, affected modules are re-tested to ensure complete resolution.

7. EXPECTED OUTCOME

By the conclusion of this phase, the system guarantees accurate input handling, reliable backend API operations, stable AI integration, smooth UI response rendering, robust error management, and consistent overall usability across diverse legal queries.

8. PHASE CONCLUSION

The Project Testing Phase verifies that LegalEase AI meets all specified operational requirements. Systematically addressing edge cases and connectivity bugs validates the end-to-end response pipeline, preparing the system for Project Documentation and Demonstration.